

Circles and circle theorems - Set 1 - Solutions

Specification MM3.2, MM3.3, M5.8, M5.9.

Answer key

Question	Answer	Topic
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Worked solutions

Question 1 CIR-01 · Circles and circle theorems · spec MM3.2

Find the radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.

Answer: B

$$(x - 3)^2 + (y + 2)^2 = 12 + 9 + 4 = 25, \text{ so } r = 5.$$

Full method - how and why

The expanded form of a circle. Complete the square in x and in y to recover the centre-radius form.

- 1 Group the terms: $(x^2 - 6x) + (y^2 + 4y) = 12$.

why: Move the constant to the right first so the completed squares have somewhere to go.

- 2 Complete each square: $x^2 - 6x = (x - 3)^2 - 9$ and $y^2 + 4y = (y + 2)^2 - 4$.

why: Halve each coefficient ($-6 \rightarrow -3$, $4 \rightarrow 2$) and subtract its square.

- 3 Substitute: $(x - 3)^2 + (y + 2)^2 - 13 = 12$, so $(x - 3)^2 + (y + 2)^2 = 25$.

why: The -9 and -4 move across to the right, **adding** to the constant.

- 4 Read off $r^2 = 25$, so $r = 5$.

why: The right-hand side is r^2 , not r .

Where it goes wrong

- Answering 25 - that is r^2 .
- Subtracting the 9 and 4 on the right instead of adding, giving $r^2 = 12$ or similar.

Takeaway $x^2 + y^2 + cx + dy + e = 0$ has centre $(-\frac{c}{2}, -\frac{d}{2})$ and $r^2 = \frac{c^2}{4} + \frac{d^2}{4} - e$.

Question 2 CIR-03 · Circles and circle theorems · spec MM3.2

The circle with centre $(2, -1)$ passes through $(5, 3)$. Find the equation of the tangent to the circle at $(5, 3)$.

Answer: B

Radius direction $(3, 4)$ has gradient $\frac{4}{3}$, so the tangent has gradient $-\frac{3}{4}$:

$$y - 3 = -\frac{3}{4}(x - 5) \Rightarrow 4y - 12 = -3x + 15 \Rightarrow 3x + 4y = 27.$$

Full method - how and why

A tangent is perpendicular to the radius at the point of contact. Find the radius gradient, invert, and write the line.

- 1 Vector from centre $(2, -1)$ to the point $(5, 3)$: $(3, 4)$, so the radius has gradient $\frac{4}{3}$.

why: Differences of coordinates give the direction; no need for the circle's equation.

- 2 The tangent is perpendicular, so its gradient is $-\frac{3}{4}$.

why: Tangent $\perp$ radius is the key circle fact here.

- 3 Line through $(5, 3)$: $y - 3 = -\frac{3}{4}(x - 5)$.

why: The tangent touches at the given point, so use it in the point-gradient form.

- 4 Clear fractions: $4y - 12 = -3x + 15 \Rightarrow 3x + 4y = 27$.

why: Multiply by 4, then collect.

Where it goes wrong

- Using the radius gradient $\frac{4}{3}$ for the tangent, giving $4x - 3y = 11$ - an offered option.
- Arithmetic slip on the constant, giving $3x + 4y = 15$.

Takeaway Tangent gradient $= -1 \div (\text{radius gradient})$; the point of contact is on both the circle and the tangent.

Question 3 CIR-04 · Circles and circle theorems · spec MM3.2

The points $(1, 7)$ and $(5, -1)$ are the ends of a diameter of a circle. Find the equation of the circle.

Answer: D

Centre is the midpoint $(3, 3)$; $\text{radius}^2 = (3 - 1)^2 + (3 - 7)^2 = 4 + 16 = 20$.

Full method - how and why

A diameter's endpoints give the centre (midpoint) and the radius (half the distance) in one move.

1 Centre = midpoint = $\left(\frac{1+5}{2}, \frac{7+(-1)}{2}\right) = (3, 3)$.

why: The centre is the midpoint of any diameter.

2 Radius²: distance from the centre to $(1, 7)$ squared = $(3 - 1)^2 + (3 - 7)^2 = 4 + 16 = 20$.

why: Going from centre to an endpoint is one radius; squaring avoids surds entirely.

3 Write the equation: $(x - 3)^2 + (y - 3)^2 = 20$.

why: The right-hand side is r^2 .

Where it goes wrong

- Using the full diameter length squared (80) as r^2 .
- Writing $r^2 = \sqrt{20}$ - the square root is already undone by squaring.

Takeaway From a diameter: centre is the midpoint, and r^2 is the squared distance from centre to either end.

Question 4 CIR-06 · Circles and circle theorems · spec MM3.3

A , B and C lie on a circle with centre O . Angle $AOB = 100^\circ$ and C lies on the major arc AB . Find angle ACB .

Answer: D

The angle at the centre is twice the angle at the circumference subtended by the same arc: $ACB = 50^\circ$.

Full method - how and why

The angle at the centre is twice the angle at the circumference on the same arc - provided you use the right arc.

1 Identify the arc: angle $AOB = 100^\circ$ at the centre stands on the **minor** arc AB .

why: The reflex angle at the centre (260°) would correspond to the other arc - which arc you use decides the answer.

2 C is on the major arc, so it sees the same minor arc AB .

why: The circumference angle must be on the opposite arc from the one it stands on.

3 Apply the theorem: angle $ACB = \frac{1}{2} \times 100^\circ = 50^\circ$.

why: Centre angle is double the circumference angle on the same arc.

Where it goes wrong

- Doubling instead of halving, giving 200° - impossible for an angle in a triangle.
- Using the reflex angle 260° and answering 130° (which would be right if C were on the minor arc).

Takeaway Angle at centre = $2 \times$ angle at circumference **on the same arc**; check which side of the chord the point lies.

Question 5 CIR-07 · Circles and circle theorems · spec MM3.3

$PQRS$ is a cyclic quadrilateral with angle $P = 3x$, angle $R = x + 40^\circ$. Find x .

Answer: C

Opposite angles of a cyclic quadrilateral sum to 180° : $4x + 40 = 180$, $x = 35^\circ$.

Full method - how and why

Cyclic quadrilateral: opposite angles are supplementary. That gives a linear equation immediately.

- 1 Identify the opposite pair: in $PQRS$ the vertices go round in order, so P and R are opposite.

why: Opposite means separated by one vertex on each side - reading the labelling order matters.

- 2 Apply the theorem: $3x + (x + 40) = 180$.

why: Opposite angles of a cyclic quadrilateral sum to 180° .

- 3 Solve: $4x = 140$, so $x = 35^\circ$.

why: Check: angles 105° and 75° sum to 180° ✓.

Where it goes wrong

- Setting the angles equal (that is a parallelogram property, not a cyclic one).
- Using 360° and getting $x = 80^\circ$.

Takeaway Cyclic quadrilateral: opposite angles sum to 180° ; adjacent ones have no such relation.

Question 6 CIR-08 · Circles and circle theorems · spec MM3.3

A chord of a circle of radius 13 is at perpendicular distance 5 from the centre. Find the length of the chord.

Answer: F

The perpendicular from the centre bisects the chord: half-length = $\sqrt{13^2 - 5^2} = 12$, so the chord is 24.

Full method - how and why

The perpendicular from the centre bisects the chord, creating a right-angled triangle with the radius as hypotenuse.

- 1 Draw the radius to one end of the chord and the perpendicular from the centre: they form a right-angled triangle.

why: The perpendicular distance, half the chord and the radius are the three sides.

- 2 Apply Pythagoras: half-chord = $\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$.

why: The radius 13 is the hypotenuse because it is the longest side.

- 3 Double it: the chord is 24.

why: The perpendicular **bisects** the chord, so the triangle gives only half of it.

Where it goes wrong

- Answering 12 by forgetting to double.
- Adding instead of subtracting inside the square root.

Takeaway Perpendicular from centre to chord: it bisects the chord, so use Pythagoras and then double.

Question 7 CIR-10 · Circles and circle theorems · spec MM3.3

TA is a tangent to a circle at A , and AB is a chord. C is a point on the circle on the opposite side of AB from T , with angle $ACB = 65^\circ$. Find angle TAB .

Answer: B

Alternate segment theorem: the angle between tangent and chord equals the angle in the alternate segment, so $TAB = 65^\circ$.

Full method - how and why

The alternate segment theorem, stated almost word for word. The work is in identifying which angle is 'in the alternate segment'.

- 1 TA is tangent at A and AB is a chord, so the angle TAB between them is the tangent-chord angle.
why: The theorem applies exactly to this configuration.
- 2 The alternate segment is the one on the other side of AB from T - which is where C lies.
why: The question places C deliberately; the angle used must be in the segment **away** from the tangent's side.
- 3 Therefore angle $TAB = \text{angle } ACB = 65^\circ$.
why: Alternate segment theorem: the tangent-chord angle equals the inscribed angle in the alternate segment.

Where it goes wrong

- Using $180^\circ - 65^\circ = 115^\circ$, which is the angle on the other side of the tangent.
- Halving to 32.5° by confusing this with the centre-circumference theorem.

Takeaway Tangent-chord angle = angle in the alternate segment - the segment on the far side of the chord.

Question 8 CIR-12 · Circles and circle theorems · spec MM3.2

The circle C has equation $x^2 + y^2 - 4x - 6y + 9 = 0$. Which of the following lines does not meet C ?

Answer: D

Centre $(2, 3)$, radius 2, so the circle spans $0 \leq x \leq 4$ and $1 \leq y \leq 5$. The line $y = 0$ is 3 from the centre, more than the radius, so it misses; $x = 0$, $x = 4$ and $y = 5$ touch and $x = 2$ passes through the centre.

Full method - how and why

Which line misses the circle: compare each line's distance from the centre with the radius. Horizontal and vertical lines make this trivial.

- 1 Complete the square: $(x - 2)^2 + (y - 3)^2 = -9 + 4 + 9 = 4$, so centre $(2, 3)$ and radius 2.
why: Always convert to centre-radius form first.
- 2 The circle therefore spans x from 0 to 4 and y from 1 to 5.
why: Centre $\pm$ radius in each direction gives the bounding box, whose edges are tangent lines.
- 3 Check each line: $x = 0$, $x = 4$ and $y = 5$ are tangents (distance exactly 2); $x = 2$ passes through the centre.
why: For a vertical line $x = k$, the distance from the centre is $|k - 2|$.
- 4 $y = 0$ is at distance $|0 - 3| = 3 > 2$, so it misses the circle.
why: Distance greater than the radius means no intersection.

Where it goes wrong

- Assuming a tangent line 'does not meet' the circle - touching counts as meeting.
- Getting $r^2 = 9$ by mishandling the $+9$ constant.

Takeaway For horizontal/vertical lines just compare the centre coordinate with the radius; tangency still counts as meeting.

Question 9 CIR-13 · Circles and circle theorems · spec MM3.2

The circle $x^2 + y^2 = r^2$ and the line $3x + 4y = 20$ have exactly one point in common. Find r .

Answer: B

Distance from origin to the line $= \frac{20}{\sqrt{9+16}} = 4$. Tangency means $r = 4$.

Full method - how and why

One common point means tangency, so the radius equals the distance from the centre to the line.

- 1 The circle is centred at the origin, so compute the distance from $(0, 0)$ to $3x + 4y = 20$.

why: Centre-at-origin makes the distance formula especially clean.

- 2 Distance $= \frac{|3(0) + 4(0) - 20|}{\sqrt{3^2 + 4^2}} = \frac{20}{5} = 4$.

why: $\sqrt{9+16} = 5$ the familiar 3-4-5 triple.

- 3 Tangency requires r equal to this distance: $r = 4$.

why: If r were larger the line would cut the circle twice; smaller, and it would miss.

Where it goes wrong

- Dividing by $\sqrt{20}$ or by 7 instead of 5.
- Answering 20 or $\sqrt{20}$ by stopping too early.

Takeaway Distance from the origin to $ax + by = c$ is $\frac{|c|}{\sqrt{a^2 + b^2}}$; tangency sets it equal to r .

Question 10 CIR-14 · Circles and circle theorems · spec MM3.3

AB is a diameter of a circle and C is a point on the circle with $AC = 6$ and $BC = 8$. Find the radius of the circle.

Answer: E

The angle in a semicircle is a right angle, so $AB = \sqrt{36 + 64} = 10$ and the radius is 5.

Full method - how and why

The angle in a semicircle is a right angle, so AB is the hypotenuse of a right-angled triangle.

- 1 AB is a diameter and C is on the circle, so angle $ACB = 90^\circ$.

why: The angle in a semicircle is a right angle - the key theorem here.

- 2 Apply Pythagoras: $AB = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$.

why: The right angle is at C , so the diameter AB is the hypotenuse.

- 3 Radius $= \frac{10}{2} = 5$.

why: The question asks for the radius, not the diameter.

Where it goes wrong

- Answering 10 (the diameter).
- Assuming the right angle is at A or B .

Takeaway A triangle inscribed with one side a diameter is right-angled at the third vertex.

Question 11 CIR-15 · Circles and circle theorems · spec MM3.2

A circle passes through $(0, 0)$, $(6, 0)$ and $(0, 8)$. Find the coordinates of its centre.

Answer: E

The triangle has a right angle at the origin, so $(6, 0)$ - $(0, 8)$ is a diameter; the centre is its midpoint $(3, 4)$.

Full method - how and why

Three points determine a circle, but here two of the chords are along the axes - and the right angle at the origin gives the centre instantly.

- 1 Notice the triangle $(0, 0)$, $(6, 0)$, $(0, 8)$ has a right angle at the origin.
why: The two legs lie along the axes, so the angle between them is 90° .
- 2 A right angle inscribed in a circle stands on a diameter, so the hypotenuse from $(6, 0)$ to $(0, 8)$ is a diameter.
why: This is the converse of the angle-in-a-semicircle theorem.
- 3 The centre is the midpoint of that diameter: $\left(\frac{6+0}{2}, \frac{0+8}{2}\right) = (3, 4)$.
why: Centre is the midpoint of any diameter.

Where it goes wrong

- Averaging all three points to get $(2, \frac{8}{3})$ - that is the centroid, not the circumcentre.
- Taking the midpoint of the wrong pair, e.g. $(3, 0)$.

Takeaway If three given points form a right angle, the hypotenuse is a diameter and its midpoint is the centre.

Question 12 X2-10 · Circles and circle theorems · spec MM3.2

The circle C has centre (a, b) and radius r . P : ' $r > |b|$ '. Q : ' C crosses the x -axis at two distinct points'. Which one of the following is true?

Answer: A

The distance from the centre to the x -axis is $|b|$; the circle cuts the axis twice exactly when this is less than r .

Full method - how and why

Translate the geometry into a distance comparison: circle versus line depends on centre-to-line distance against the radius.

- 1 The perpendicular distance from the centre (a, b) to the x -axis is $|b|$.
why: The x -axis is $y = 0$, so the vertical distance is the size of the y -coordinate.
- 2 If $|b| < r$ the circle reaches across the axis and cuts it twice; if $|b| = r$ it touches once; if $|b| > r$ it misses entirely.
why: Three cases, decided by comparing distance with radius.
- 3 So 'two distinct points' happens exactly when $r > |b|$ - the two statements are equivalent.
why: Necessary and sufficient.

Where it goes wrong

- Writing the condition as $r > b$, which fails for a centre below the axis (b negative). The modulus matters.
- Including the tangent case $r = |b|$, which gives one point, not two.

Takeaway Circle versus line: compare the perpendicular distance from the centre with r - less gives two points, equal one, greater none.